Track 36: BIM & Structural Engineering

June 2026

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Department: Civil Engineering | **MR ROKESH TECHNOLOGY**

> ***Skill path:*** *Revit Architecture → Structural Modeling → STAAD.Pro → Analysis → BIM Coordination → Clash Detection*

Related: Track 35 — AutoCAD & Civil Design | Department Map

Track Overview

Who this is for: Civil/structural engineering students targeting BIM modeler and structural analyst roles.

Prerequisites: Basic AutoCAD or engineering drawing; Track 35 recommended first.

Tools: Revit, STAAD.Pro (or ETABS intro), Navisworks (clash detection intro), AutoCAD for details.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	BIM awareness	Revit architectural model
1 Month	80–100	Junior BIM modeler	Structural model + basic analysis
3 Months	250–300	Internship-ready	Full BIM + STAAD capstone
6 Months	500–600	BIM coordinator ready	Multi-discipline BIM portfolio

2-Week Crash Course

Days	Topics	Deliverable
D1–2	BIM concepts, Revit interface, levels, grids	Revit project setup
D3–4	Walls, floors, roofs, doors, windows	Architectural model
D5–6	Structural elements: beams, columns, foundations	Structural Revit model
D7–8	STAAD.Pro: geometry, loads, support conditions	Analysis model
D9–10	Results interpretation, capstone	**BIM model + analysis report**

1-Month Foundation

Week	Topics	Project
W1	Revit architecture modeling	G+1 building BIM model
W2	Revit Structure, rebar detailing intro	Structural model
W3	STAAD.Pro: DL, LL, wind/seismic intro	Analysis + design report
W4	Documentation, schedules, capstone	**Complete BIM + STAAD package**

3-Month Job Ready

Month	Topics	Projects
M1	Advanced Revit, families, templates	Custom family library
M2	STAAD advanced: combinations, steel/concrete design	Multi-storey analysis
M3	Navisworks clash detection, 4D intro, capstone	**Multi-discipline BIM capstone**

6-Month Professional

Add: ETABS comparison, construction documentation, BOQ from BIM, 5-project portfolio, client presentation skills.

Assessment Rubric

Weekly Quiz 20% | Model/analysis labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Validate STAAD results against hand calculations for simple beams. Export coordinated models for clash review in 3-month+ tiers.